

EDUCATION

UNIVERSITY OF WISCONSIN-MADISON

Expected May 2028

Astronomy Ph.D.

National Science Foundation Graduate Research Fellow

MSc. conferred May 2025

UNIVERSITY OF PITTSBURGH

Conferred May 2022

B.S. in Physics and Astronomy

Summa cum Laude

Dean's List all semesters

RESEARCH EXPERIENCE

Robust Evolution of the Mass–Metallicity Relation

2025–

Department of Astronomy, University of Wisconsin-Madison, Madison, WI

Forward modeled observed mass metallicity relations across cosmic time, accounting for selection effects and observational biases.

RUBIES Corrected Mass–Metallicity Relation

2023–2025

Department of Astronomy, University of Wisconsin-Madison, Madison, WI

Characterized mass–metallicity relation of massive galaxies in RUBIES at $z \sim 4$ and corrected for survey selection effects and biases, in collaboration with Prof. Michael Maseda.

LEGA-C Mass–Metallicity Relation

2019–2024

Department of Physics and Astronomy, University of Pittsburgh, Pittsburgh, PA

Researched the gas phase mass-metallicity relation of massive star-forming galaxies in the LEGA-C dataset at $z \sim 0.8$, in collaboration with Prof. Brett Andrews and Prof. Rachel Bezanson.

ATLAS Collision Detection

2019

Department of Physics and Astronomy, University of Pittsburgh, Pittsburgh, PA

Worked with ROOT and C macros to analyze particle collisions in the ATLAS Detector at the Large Hadron Collider, in collaboration with Prof. Tae Min Hong.

OUTREACH AND ACADEMIC SERVICE

ACORNS (Activities for Community OutReach iN STEM)

2024–

Madison, WI

Created an outreach initiative to connect the UW-Madison astronomy department to community centers throughout Dane County. This involves elementary-age students at these centers visiting the astronomy department on campus, and graduate students giving hands-on demonstrations at the centers. In 2025, expanded this program to include biology, chemistry, and

physics departments at UW-Madison.

Climate Committee Graduate Student Representative

2023-2025

Madison, WI

Was elected as one of two graduate student representatives on the UW-Madison Astronomy Department Climate Committee. Primary role included the creation, distribution, and analysis of a department-wide climate survey. Chaired Climate Committee 2024-2025.

Secretary of Society of Physics Students Chapter

2021-2022

Pittsburgh, PA

Was elected secretary of SPS chapter. In this role, I created a “physics survival guide” that included a set of detailed instructions on how to begin coding in Python, including introductory scripts with information relevant to those doing research in astronomy.

Undergraduate Mentoring in Society of Physics Students

2020-2021

Pittsburgh, PA

Mentored two undergraduate students through the University of Pittsburgh’s chapter of SPS while an upper-year undergraduate. This experience helped me to develop my mentoring style—such as creating a dynamic, two-way relationship that encourages feedback and allows me to be responsive to the mentee’s changing needs.

Department Equity and Inclusion Committee Member

2020-2021

Pittsburgh, PA

Was elected to the Department’s Equity and Inclusion Committee. Throughout my time on this committee, I learned what it takes for department policy to be changed, ways to present ideas for said change so that it is palatable to the many different strata of a department, and how best to use institutional knowledge to the advantage of equity work.

ExpERIENCE Children’s Museum Education Assistant

2016-2018

Erie, PA

Worked as an education assistant at the ExpERIENCE Children’s Museum. Helped to run science-themed summer camps for groups of size ~20 ranging from ages 5–10.

TEACHING EXPERIENCE

Teaching Assistant for Undergraduate Astronomy Course

Fall 2023

University of Wisconsin-Madison Teaching assistant for undergraduate astronomy course geared at non-majors: “The Dark Side of the Universe”. Topics included black holes, dark matter, and dark energy. In this role I led review sessions and hosted office hours.

Teaching in the College Classroom: DELTA Course

Fall 2023

University of Wisconsin-Madison Took course on equitable and sustainable teaching practices in the college classroom. Designed curriculum, assignments, and lesson plans. Performed practice teaching.

CONFERENCES AND WORKSHOPS

54th Saas-Fee Advanced Course: Galaxies and Black Holes in the First Billion Years with JWST January 2025

Saas-Fee, Switzerland

Week-long course for Ph.D. students focused on the theory and observation of early galaxy evolution in the era of JWST.

RUBIES Collaboration Meeting July 2024

Madison, WI, USA

Collaboration meeting for the Red Unknowns: Bright Infrared Emitters Survey. Presented work on the RUBIES mass–metallicity relation.

IMPRS “Unraveling Galaxy Evolution with JWST” Summer School September 2023

Heidelberg, Germany

Attended week-long workshop hosted by Max Planck Institute for Astronomy. Lectures and hands-on lessons covered JWST data reduction, proposal writing, and the physics of nearby and distant galaxies.

LEGA-C Collaboration Meeting June 2022

Bruges, Belgium

Attended collaboration meeting for Large Early Galaxy Astrophysics Census (LEGA-C) survey. Presented on my work on the LEGA-C mass–metallicity relation.

Astrophysics Boot Camp May 2021

Pittsburgh, PA, USA

Contributed a talk on advanced NumPy techniques to week-long astrophysics boot camp hosted by University of Pittsburgh as an undergraduate. This “camp” taught astronomical coding subjects such as astronomical software libraries and accessible plotting practices.

HONORS AND AWARDS

National Science Foundation Graduate Research Fellowship	2024-2027
Wisconsin Space Grant Consortium (x3)	2024-2027
Halliday-Resnick Award for Excellence in Undergraduate Research	2022
Julia Thompson Award for Excellence in Undergraduate Writing	2022
Pennsylvania Space Grant Consortium Scholarship (x7)	2019-2022

PUBLICATIONS

- [1] Nikko J. Cleri, **Lewis, Zach J.**, Joel Leja, Jakob M. Helton, Emilie Burnham, Olivia Curtis, Anna de Graaff, Michaela Hirschmann, Harley Katz, Michael V. Maseda, Ian McConachie, Adele Plat, and Lucie Scharre. “RUBIES: The Evolution of the Ionization Parameter from $0 < z < 9$ ”. In: *arXiv e-prints*, arXiv:2605.30410 (May 2026), arXiv:2605.30410. DOI: 10.48550/arXiv.2605.30410. arXiv: 2605.30410 [astro-ph.GA].

- [2] Nikko J. Cleri, Grace M. Olivier, Bren E. Backhaus, Joel Leja, Casey Papovich, Jonathan R. Trump, Pablo Arrabal Haro, Véronique Buat, Denis Burgarella, Emilie Burnham, Antonello Calabrò, Jonathan H. Cohn, Justin W. Cole, Kelcey Davis, Mark Dickinson, Steven L. Finkelstein, Ray Garner III, Michaela Hirschmann, Weida Hu, Taylor A. Hutchison, Dale D. Kocevski, Anton M. Koekemoer, Rebecca L. Larson, **Lewis, Zach J.**, Michael V. Maseda, Lise-Marie Seillé, and Raymond C. Simons. “*Optical Strong Line Ratios Cannot Distinguish between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities*”. In: 994.2, 146 (Dec. 2025), p. 146. DOI: 10.3847/1538-4357/ae0f17. arXiv: 2506.21660 [astro-ph.GA].
- [3] Anna de Graaff, Gabriel Brammer, Andrea Weibel, **Lewis, Zach**, Michael V. Maseda, Pascal A. Oesch, Rachel Bezanson, Leindert A. Boogaard, Nikko J. Cleri, Olivia R. Cooper, Rashmi Gottumukkala, Jenny E. Greene, Michaela Hirschmann, Raphael E. Hviding, Harley Katz, Ivo Labbé, Joel Leja, Jorryt Matthee, Ian McConachie, Tim B. Miller, Rohan P. Naidu, Sedona H. Price, Hans-Walter Rix, David J. Setton, Katherine A. Suess, Bingjie Wang, Katherine E. Whitaker, and Christina C. Williams. “*RUBIES: A complete census of the bright and red distant Universe with JWST/NIRSpec*”. In: 697, A189 (May 2025), A189. DOI: 10.1051/0004-6361/202452186. arXiv: 2409.05948 [astro-ph.GA].
- [4] **Lewis, Zach**, Michael V. Maseda, Anna de Graaff, Joel Leja, Bingjie Wang, Hans-Walter Rix, Ian McConachie, Nikko J. Cleri, Rachel Bezanson, Leindert A. Boogaard, Gabriel Brammer, Jenny E. Greene, Michaela Hirschmann, Harley Katz, Ivo Labbe, Jorryt Matthee, Tim B. Miller, Rohan P. Naidu, Pascal A. Oesch, David J. Setton, Katherine A. Suess, Andrea Weibel, Katherine E. Whitaker, and Christina C. Williams. “*The Mass-Metallicity Relation and its Observational Effects at $z \sim 3-6$* ”. In: *arXiv e-prints*, arXiv:2512.03134 (Dec. 2025), arXiv:2512.03134. DOI: 10.48550/arXiv.2512.03134. arXiv: 2512.03134 [astro-ph.CO].
- [5] **Lewis, Zach**, Michael V. Maseda, Anna de Graaff, Joel Leja, Bingjie Wang, Hans-Walter Rix, Ian McConachie, Nikko J. Cleri, Rachel Bezanson, Leindert A. Boogaard, Gabriel Brammer, Jenny E. Greene, Michaela Hirschmann, Harley Katz, Ivo Labbé, Jorryt Matthee, Tim B. Miller, Rohan P. Naidu, Pascal A. Oesch, David J. Setton, Katherine A. Suess, Andrea Weibel, Katherine E. Whitaker, and Christina C. Williams. “*The Mass-Metallicity Relation and Its Observational Effects at $z \sim 3-6$* ”. In: 1005.2, 159 (July 2026), p. 159. DOI: 10.3847/1538-4357/ae7bfc. arXiv: 2512.03134 [astro-ph.CO].
- [6] **Lewis, Zach J.**, Brett H. Andrews, Rachel Bezanson, Michael Maseda, Eric F. Bell, Romeel Davé, Francesco D’Eugenio, Marijn Franx, Anna Gallazzi, Anna de Graaff, Yasha Kaushal, Angelos Nersesian, Jeffrey A. Newman, Arjen van der Wel, and Po-Feng Wu. “*The Gas-phase Mass–Metallicity Relation for Massive Galaxies at $z \sim 0.7$ with the LEGA-C Survey*”. In: 964.1, 59 (Mar. 2024), p. 59. DOI: 10.3847/1538-4357/ad250c. arXiv: 2304.12343 [astro-ph.GA].
- [7] Michael V. Maseda, **Lewis, Zach**, Jorryt Matthee, Joseph F. Hennawi, Leindert Boogaard, Anna Feltre, Themiya Nanayakkara, Roland Bacon, Amy Barger, Jarle Brinchmann, Marijn Franx, Takuya Hashimoto, Hanae Inami, Haruka Kusakabe, Floriane Leclercq, Lucie Rowland, Anthony J. Taylor, Christy Tremonti, Tanya Urrutia, Joop Schaye, Charlotte Simmonds, and Eloise Vitte. “*JWST/NIRSpec Measurements of Extremely Low Metallicities in High Equivalent Width $\text{Ly}\alpha$ Emitters*”. In: 956.1, 11 (Oct. 2023), p. 11. DOI: 10.3847/1538-4357/acf12b. arXiv: 2304.08511 [astro-ph.GA].

- [8] Owen S. Matthews Acuña, Christy A. Tremonti, Kyle B. Westfall, Shea DeFour-Remy, Aleksandar M. Diamond-Stanic, **Lewis, Zach J.**, Britt Lundgren, Drake Miller III, and Lizhou Sha. “*Precision Spectroscopy for 1.9 Million Galaxies from SDSS-IV: Improved Spectral Measurements and Catalogs for eBOSS*”. In: *arXiv e-prints*, arXiv:2512.18076 (Dec. 2025), arXiv:2512.18076. DOI: 10 . 48550 / arXiv . 2512 . 18076. arXiv: 2512 . 18076 [astro-ph.GA].
- [9] Ian McConachie, Jacqueline Antwi-Danso, Wenjun Chang, M. C. Cooper, Adit Edward, Ben Forrest, Percy Gomez, Han Lei, **Lewis, Zach J.**, Danilo Marchesini, Michael V. Maseda, Adam Muzzin, Allison Noble, Stephanie M. Urbano Stawinski, Tracy Webb, Gillian Wilson, and M. E. Wisz. “*Excavating The Ruins: an Ancient $z = 2.675$ Galaxy Which Formed in the First 500 Myr*”. In: *arXiv e-prints*, arXiv:2508.05752 (Aug. 2025), arXiv:2508.05752. DOI: 10 . 48550 / arXiv . 2508 . 05752. arXiv: 2508 . 05752 [astro-ph.GA].
- [10] Ian McConachie, Anna de Graaff, Michael V. Maseda, Joel Leja, Yunchong Zhang, David J. Setton, Rachel Bezanson, Leindert A. Boogaard, Gabriel Brammer, Nikko J. Cleri, Olivia R. Cooper, Karl Glazebrook, Rashmi Gottumukkala, Jenny E. Greene, Andy D. Goulding, Michaela Hirschmann, Ivo Labbe, **Lewis, Zach**, Jorryt Matthee, Tim B. Miller, Rohan P. Naidu, Pascal A. Oesch, Sedona H. Price, Themiya Nanayakkara, Katherine A. Suess, Bingjie Wang, Katherine E. Whitaker, and Christina C. Williams. “*Where Galaxies Go to Die: The Environments of Massive Quiescent Galaxies at $3 < z < 5$* ”. In: *arXiv e-prints*, arXiv:2510.25024 (Oct. 2025), arXiv:2510.25024. DOI: 10 . 48550 / arXiv . 2510 . 25024. arXiv: 2510 . 25024 [astro-ph.GA].